

Answer **all** questions.

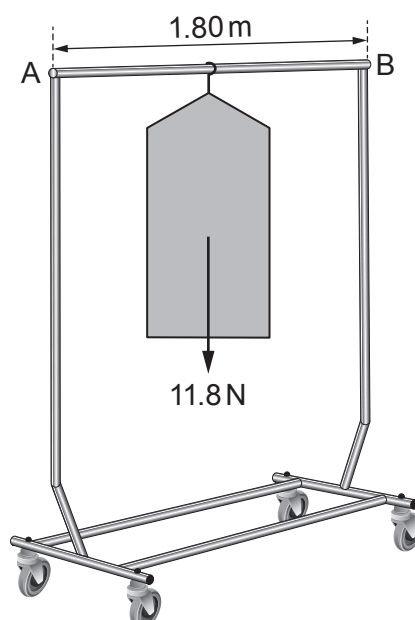
1. (a) State the conditions needed for a body to remain in equilibrium. [2]

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- (b) The diagram shows a simple clothes carrier. It consists of a uniform horizontal beam (AB) of length 1.80 m resting on two vertical rods.



When the clothes carrier supports a coat weighing 11.8 N **at its centre**, a force of 8.0 N is exerted on each rod. Show that the weight, W , of the horizontal beam is approximately 4 N.

[2]

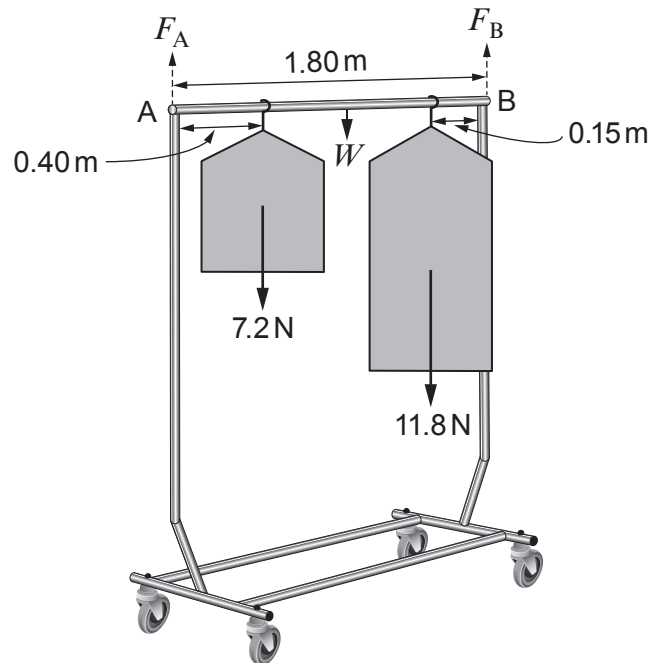
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- (c) The coat is now moved to a new position and a jacket weighing 7.2 N is hung from the beam, all of which is shown below.



Calculate forces F_A and F_B .

[5]

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